MAHATMA GANDHI UNIVERSITY, KOTTAYAM

ANSWER KEY

Fifth Semester – October 2026 Examination
MG5DSEFTQ303 – TECHNOLOGY OF SPICES

(2024 Admission Onwards)

PART A – SHORT ANSWER QUESTIONS

Answer any 10 questions. Each question carries 2 marks.

1. Name two spices that possess antimicrobial properties.

Clove and cinnamon possess antimicrobial properties.
Other acceptable examples include garlic, turmeric, black pepper and oregano.

2. Name any two natural drying methods used for spices.

Sun drying.
Shade drying (air drying).

3. List the major steps involved in the processing of nutmeg.

Harvesting of mature fruits, followed by removal of the outer fleshy pericarp.
Separation of the mace (aril) from the nut; drying of mace separately.
Drying of the nut, cracking the shell, removal of the kernel, grading and packing.

4. Recall the gases used in fumigation of spices.

Commonly cited fumigants include ethylene oxide and, historically, methyl bromide.
Use is subject to applicable food-safety regulations and restrictions.

5. Define cleaning of spices.

Cleaning is the unit operation of removing foreign and unwanted materials such as dust, soil, stones, stalks, insects and other extraneous matter from spices.
It improves purity, safety and market quality.

6. List two applications of spice oleoresins in the food industry.

Used as flavouring ingredients in processed foods such as soups, sauces, meat products, snacks and seasonings.
Used in spice blends, marinades and convenience foods to provide consistent flavour and colour.

7. Define ginger.

Ginger is the underground stem (rhizome) of Zingiber officinale, a herbaceous plant belonging to the family Zingiberaceae.
It is used as a spice for its pungent taste and characteristic aroma.

8. What is black pepper?

Black pepper is the dried, usually unripe fruit (drupe/peppercorn) of Piper nigrum.
It is known for its pungency, mainly due to the alkaloid piperine.

9. Differentiate between ethylene oxide treatment and conventional thermal treatment.

Ethylene oxide treatment is a chemical fumigation process used to reduce microbial contamination; it can be carried out at relatively low temperatures.

Thermal treatment uses heat (for example, steam or hot air) to destroy microorganisms.

Ethylene oxide requires strict control of residues, aeration and worker safety; heat treatment may cause loss of volatile aroma compounds or colour changes.

10. Mention two methods of drying cardamom.

Traditional sun drying.

Artificial/mechanical drying using hot air or a cardamom drier (including flue-pipe or tray drying systems).

11. State one important quality parameter of spice oleoresins.

One important parameter is the content of the characteristic active principle or marker compound (for example, pungency/colour value).

Other acceptable parameters include volatile oil content, colour, moisture, solvent residue and sensory quality.

12. Name the major colouring principle present in turmeric.

Curcumin is the major colouring principle in turmeric.

It belongs to the curcuminoid group of yellow pigments.

PART B – SHORT ESSAY QUESTIONS

Answer any 6 questions. Each question carries 5 marks.

13. Describe the different methods of drying cardamom.

Sun drying: Capsules are spread in a thin layer on clean drying floors or mats and turned periodically. It is economical but depends on weather and may result in uneven drying or colour loss.

Flue-pipe drying: Capsules are dried in a chamber using heated air circulated through flue pipes. The produce is not intended to come into direct contact with smoke or combustion gases.

Hot-air/mechanical drying: Controlled heated air is passed through trays or a drying chamber. Temperature and airflow can be regulated, giving more uniform and faster drying.

Tray or cabinet drying: Cardamom is spread in shallow layers on trays and dried under controlled conditions.

After drying, capsules are cooled, cleaned, graded and packed. Gentle drying helps retain the green colour, aroma and essential oil.

14. Explain the major factors affecting the rate of drying of spices.

Temperature of drying air: Higher temperature generally increases the rate of moisture evaporation, but excessive heat can damage colour and volatile oils.

Relative humidity: Low relative humidity increases the moisture-removing capacity of air; high humidity slows drying.

Air velocity and circulation: Faster, uniform airflow removes moist air from around the product and improves drying.

Size, shape and surface area: Small pieces and thin layers dry faster because moisture has a shorter path to move and more surface is exposed.

Initial moisture content and composition: High initial moisture generally requires longer drying; oil-rich or dense tissues may dry more slowly.

Loading depth and arrangement: Thick layers and overcrowding restrict airflow and slow drying.

Nature of the product and pretreatment: Cutting, splitting or blanching (where suitable) can alter moisture movement and drying time.

15. Describe the volatility, aroma and flavour characteristics of spice oils.

Spice oils are mixtures of volatile aromatic compounds, commonly including terpenes, terpenoids, aldehydes, phenols, alcohols and esters.

Volatility: These compounds evaporate readily, particularly during heating and prolonged exposure to air. Volatility varies with molecular structure and boiling point.

Aroma: The characteristic smell of each spice is produced by its specific combination and proportion of volatile compounds. Examples include eugenol in clove and cinnamaldehyde in cinnamon bark oil.

Flavour: Spice oils contribute aroma and taste sensations such as warm, sweet, pungent, fresh or herbal notes. Some compounds also contribute to pungency.

Storage: Heat, oxygen, light and moisture can cause evaporation or oxidation, reducing aroma quality. Oils should be stored in tightly closed, suitable containers away from heat and light.

16. Differentiate between conventional extraction and supercritical carbon dioxide extraction of spice components.

Principle: Conventional extraction uses liquid solvents (such as food-grade ethanol or other permitted solvents) to dissolve target compounds. Supercritical CO_2 extraction uses carbon dioxide above its critical temperature and pressure, where it has solvent-like properties.

Operating conditions: Conventional extraction is generally carried out at ordinary or moderately elevated temperatures and solvent pressure. Supercritical extraction requires high-pressure equipment and controlled temperature.

Selectivity: Conventional extraction selectivity depends on solvent polarity and process conditions. Supercritical CO_2 selectivity can be adjusted by changing pressure and temperature; a co-solvent may be used for more polar compounds.

Product quality: Supercritical CO_2 operates at relatively low temperatures and leaves no CO_2 residue after depressurisation, helping retain heat-sensitive aroma compounds. Conventional extraction may leave solvent residues unless adequately removed.

Cost and limitations: Conventional systems are usually simpler and less expensive but require solvent recovery and residue control. Supercritical systems have high capital and operating costs but can produce high-quality extracts.

17. Differentiate between flower spices and bud spices.

Flower spices are obtained from the flower or a part of an opened flower. Examples include saffron (dried stigmas) and rose (petals, where used as a spice/flavouring).

Bud spices are obtained from unopened flower buds. The principal example is clove, which consists of the dried unopened flower bud of Syzygium aromaticum.

Flower spices may include delicate reproductive parts such as stigma or petals, whereas bud spices are harvested before the flower opens.

Their harvesting stage, morphology, processing and characteristic aroma differ.

18. Differentiate between the processing methods of saffron and cloves, highlighting their major processing operations.

Saffron: Flowers of Crocus sativus are harvested by hand, usually early in the day. The red stigmas are carefully separated from the flowers.

The separated stigmas are dried under controlled conditions to develop the characteristic colour, aroma and flavour, then cleaned, graded and packed in moisture- and light-protective packaging.

Cloves: Unopened flower buds of Syzygium aromaticum are harvested when they reach the appropriate stage and show a characteristic colour change.

The stalks are separated as required, and buds are dried by sun or artificial drying until they become brown and sufficiently dry. They are then cleaned, graded and packed.

Key difference: Saffron processing centres on delicate hand separation and careful drying of stigmas, whereas clove processing involves harvesting and drying whole unopened flower buds.

19. Explain the advantages and limitations of irradiation treatment of spices.

Advantages: Irradiation reduces microbial load, including pathogenic and spoilage microorganisms, and can control insects and pests.

It can be carried out with limited temperature rise, helping preserve heat-sensitive flavour and aroma better than some heat treatments.

It can treat packaged products, reduce the need for chemical fumigants and extend storage life when combined with good packaging and hygiene.

Limitations: Installation and operation require specialised, costly equipment, trained personnel and regulatory approval.

The treatment does not replace good manufacturing practices and may not destroy all microbial spores under every condition.

Excessive or unsuitable doses may affect sensory or chemical quality. Labelling and dose limits must comply with applicable regulations.

20. List the advantages of plastic packaging for spices.

Lightweight, flexible and convenient to handle and transport.

Can provide a good barrier to moisture and, depending on the polymer and laminate, oxygen and aroma loss.

Helps protect spices from contamination, dust, insects and handling damage.

Can be heat-sealed to form secure packs and is available in different sizes and forms.

Transparent films allow product visibility; printed films provide branding and product information.

Generally economical and suitable for automated packaging. Barrier performance depends on the type and thickness of plastic used.

PART C – LONG ESSAY QUESTIONS

Answer any 2 questions. Each question carries 10 marks.

21. Explain the relationship between the chemical composition of spices and their functional properties in food preservation, flavouring and health promotion.

Spices contain essential oils, phenolic compounds, alkaloids, pigments, resins, pungent principles and other phytochemicals. Their composition determines their characteristic sensory and functional properties.

Food preservation – antimicrobial action: Certain phenolic and aromatic compounds inhibit bacteria, yeasts and moulds. Examples include eugenol in clove, cinnamaldehyde in cinnamon and compounds in garlic. Their effectiveness depends on concentration, food composition and the target microorganism.

Food preservation – antioxidant action: Phenolic compounds and pigments can scavenge free radicals or slow oxidation of fats, helping delay rancidity and quality deterioration. Examples include curcuminoids in turmeric and phenolic compounds in several herbs and spices.

Flavouring: Volatile essential-oil components provide characteristic aroma. Examples include eugenol (clove), cinnamaldehyde (cinnamon), menthol (mint), and terpenes such as limonene (citrus peel).

Taste and pungency: Piperine gives black pepper its pungency; gingerols and shogaols contribute to ginger pungency; capsaicinoids give chilli its heat.

Colour: Curcumin gives turmeric its yellow colour; carotenoid pigments contribute red or orange colour in chilli and paprika.

Health-related properties: Several spice constituents have been studied for antioxidant, anti-inflammatory, digestive and other biological activities. For example, curcuminoids, gingerols and capsaicin have been investigated for such effects.

The health effects of spices depend on dose, bioavailability and the overall diet. Spices are foods or food ingredients and should not be presented as substitutes for medical treatment.

Conclusion: The chemical composition of each spice determines its aroma, taste, colour, antimicrobial and antioxidant potential, making spices valuable in food formulation and quality preservation.

22. Describe the various factors affecting the storage stability and shelf life of spices.

Moisture content and water activity: High moisture encourages mould growth, insect activity, caking and quality deterioration. Spices should be dried to safe moisture levels and protected from moisture uptake.

Temperature: High storage temperatures accelerate loss of volatile oils, oxidation, colour changes and deterioration. Cool storage helps preserve quality.

Relative humidity: High humidity can cause moisture absorption and mould growth; storage areas should be dry and well ventilated.

Light: Light can degrade pigments and sensitive flavour compounds. Opaque or light-protective packaging reduces exposure.

Oxygen: Oxygen promotes oxidation of essential oils, pigments and other constituents. Airtight packaging and suitable oxygen-barrier materials help limit deterioration.

Packaging material: Packaging should protect against moisture, oxygen, light, insects and external odours. Barrier properties and seal integrity are important.

Form of spice: Whole spices generally retain volatile oils and flavour longer than ground spices because grinding increases exposed surface area.

Initial quality and microbial load: Proper harvesting, cleaning, drying and hygienic handling reduce contamination and improve keeping quality.

Insects and pests: Insect infestation causes physical damage, contamination and loss of market value. Sanitation, pest control and suitable packaging are necessary.
Storage conditions and handling: Clean, dry warehouses, pallets, protection from walls/floors, stock rotation (FIFO/FEFO) and avoidance of strong-smelling materials help maintain quality.
Conclusion: Shelf life is maintained by controlling moisture, temperature, oxygen, light, pests and contamination, together with appropriate packaging and good storage practices.

23. Explain the mechanical methods of chilli drying in detail.

Mechanical drying uses equipment to supply controlled heat and airflow, reducing dependence on sunlight and weather.
Tray/cabinet drying: Clean chillies are spread in a uniform, shallow layer on perforated trays. Heated air is circulated through the cabinet; trays may be rotated or rearranged to improve uniformity.
Hot-air or batch drying: Chillies are loaded into a drying chamber and heated air is passed over or through the produce. Temperature, air velocity and drying time are controlled to reach the required final moisture.
Continuous-belt drying: Chillies move on a perforated conveyor through zones of heated air. Residence time, belt speed and air conditions are adjusted to obtain uniform drying in larger-scale operations.
Fluidised-bed drying: In suitable systems and product sizes, upward-moving hot air suspends or agitates the product, improving heat and mass transfer. It is less commonly used for whole large chillies than for smaller particles or processed materials.
Pre-drying operations: Harvesting at proper maturity, sorting, removal of damaged pods and cleaning are carried out. Chillies may be washed only when the process includes adequate subsequent drying.
Process control: Excessively high temperature can cause colour degradation, loss of pungency/aroma and surface hardening. Air temperature, humidity, airflow, loading depth and drying time must be controlled.
Post-drying: Dried chillies are cooled, cleaned, graded and packed in moisture-resistant packaging. They should be stored in a cool, dry place away from light and pests.
Advantages: Faster and more uniform drying, reduced dependence on weather, better process control, reduced contamination risk and suitability for commercial-scale production.
Limitations: Higher equipment and energy costs; poor control may cause uneven drying, colour loss or deterioration of volatile constituents.

24. Compare propylene oxide treatment with ethylene oxide and steam sterilization for spice decontamination.

Propylene oxide (PO): A chemical fumigant used to reduce microbial contamination in spices. It reacts with cellular constituents and can be effective at relatively low temperatures. Residues and reaction products must be controlled, and use is subject to regulatory approval.

Ethylene oxide (EtO): A gaseous chemical sterilant/fumigant with strong antimicrobial activity. It can penetrate packaged or bulk products and is used at relatively low temperatures, but it presents significant worker-safety and residue concerns. Adequate aeration and strict regulatory controls are essential.
Steam sterilization: Uses moist heat to inactivate microorganisms. It avoids chemical fumigant residues and can be effective when time and temperature are properly controlled.

Effect on quality: PO and EtO may better preserve heat-sensitive volatile components than high-temperature steam, but chemical residues and safety risks are concerns. Steam may cause loss of volatile oils, changes in colour, flavour or texture if treatment is too severe.

Operational considerations: Chemical fumigation requires gas-tight equipment, monitoring, trained operators, aeration and residue testing. Steam treatment requires controlled steam exposure, followed by drying to prevent moisture-related spoilage.

Safety and regulation: The acceptability and permitted use of PO and EtO vary by jurisdiction and food category; they must not be assumed to be permitted. Applicable national regulations and maximum residue requirements must be followed. Steam is generally preferred where it can achieve the required microbial reduction without unacceptable quality loss.

Comparison summary: PO and EtO are low-temperature chemical decontamination methods with residue and occupational-safety concerns; steam is a physical heat treatment with no fumigant residue but may affect heat-sensitive spice quality.